

P4410 PROBLEM SET 3 SOLUTIONS

Shankar 4.2.1

- (1) The possible values one can obtain if L_z is measured are its eigenvalues.

$$L_z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad \text{Eigenvalues are } \{-1, 0, 1\}$$

- (2) The state in which $L_z = 1$ is the corresponding eigenvector $|L_z; 1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

In $|L_z; 1\rangle$, What is $\langle L_x \rangle$, $\langle L_x^2 \rangle$, and ΔL_x ?

$$\begin{aligned} \langle L_x \rangle_{z,1} &= \langle L_z; 1 | L_x | L_z; 1 \rangle \\ &= [1 \ 0 \ 0] \begin{bmatrix} 0 & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

$$= [1 \ 0 \ 0] \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = 0 = \langle L_x \rangle_{z,1}$$

$$\langle L_x^2 \rangle_{z,1} = [1 \ 0 \ 0] \begin{bmatrix} 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1/2 = \langle L_x^2 \rangle_{z,1}$$

$$\Delta L_x = [\langle L_z; 1 | (L_x - \langle L_x \rangle_{z,1})^2 | L_z; 1 \rangle]^{1/2}$$

$$\Delta L_x = \frac{1}{\sqrt{2}}$$

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Shankar 4.2.1 continued

(3) Find normalized eigenstates and the eigenvalues of L_x in L_z basis

$$L_x = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The eigenvalues are $\{-1, 0, 1\}$

The eigenvectors are $\left\{ \begin{pmatrix} 1 \\ -\frac{1}{\sqrt{2}} \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ \frac{1}{\sqrt{2}} \\ 1 \end{pmatrix} \right\}$

and normalizing

$$\left\{ \frac{1}{2} \begin{pmatrix} 1 \\ -\frac{1}{\sqrt{2}} \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{2} \begin{pmatrix} 1 \\ \frac{1}{\sqrt{2}} \\ 1 \end{pmatrix} \right\}$$

Writing in terms of L_z basis:

$$-1: \quad \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{\sqrt{2}}{2} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$0: \quad -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$+1: \quad \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \frac{\sqrt{2}}{2} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

4.2.1 cont'd

(4) Particle is in state $|L_z = -1\rangle \Leftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$$P(L_x = -1) = \left| \langle L_x = -1 | L_z = -1 \rangle \right|^2 = \left| \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{2}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \frac{1}{4}$$

$$P(L_x = 0) = \left| \langle L_x = 0 | L_z = -1 \rangle \right|^2 = \left| \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \frac{1}{2}$$

$$P(L_x = 1) = \left| \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{2}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \frac{1}{4}$$

(5) If $L_z^2 = 1$, $L_z = \pm 1$. If this is an ideal measurement, it leaves the particle in a state where the $L_z = 0$ component has been "knocked out".

$$|\psi\rangle \rightarrow \begin{pmatrix} 1/2 \\ 0 \\ \sqrt{1/2} \end{pmatrix}$$

But this needs to be normalized, so

$$\text{the state becomes } \begin{pmatrix} 1/2 \\ 0 \\ \sqrt{1/2} \end{pmatrix} \cdot \left[\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right]^{-1/2} = \frac{2}{\sqrt{3}} \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{1}{3}} \\ 0 \\ \sqrt{\frac{2}{3}} \end{pmatrix}$$

$$\text{Probability of this is } P(L_x = 1) + P(L_x = -1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

(6) The overall phase is irrelevant physically, but the relative phase of the three terms is important.

② Shankar 4.2.3

In x basis, $\langle p \rangle = \int dx \psi^*(x) -i\hbar \frac{d}{dx} \psi(x)$

so $-i\hbar \int dx \psi^*(x) e^{\frac{-ip_0 x}{\hbar}} \frac{d}{dx} \psi(x) e^{\frac{+ip_0 x}{\hbar}}$

$= -i\hbar \int dx \psi^*(x) e^{\frac{-ip_0 x}{\hbar}} \left[\frac{+ip_0}{\hbar} \psi(x) e^{\frac{+ip_0 x}{\hbar}} + e^{\frac{+ip_0 x}{\hbar}} \frac{d}{dx} \psi(x) \right]$

$= -i\hbar \left[\int dx \left(\frac{ip_0}{\hbar} e^{\frac{-ip_0 x}{\hbar}} \psi^*(x) e^{\frac{+ip_0 x}{\hbar}} \psi(x) \right) + \int dx \left(\psi^*(x) e^{\frac{-ip_0 x}{\hbar}} e^{\frac{+ip_0 x}{\hbar}} \frac{d}{dx} \psi(x) \right) \right]$

$= p_0 \underbrace{\int dx \psi^*(x) \psi(x)}_{=1} - i\hbar \underbrace{\int dx \psi^*(x) \frac{d}{dx} \psi(x)}_{=\langle p \rangle}$

$= p_0 + \langle p \rangle \quad \text{QED.}$

③ Shankar 5.1.1

$E = \frac{p^2}{2m} \Rightarrow p = \alpha \sqrt{2mE} \quad ; \text{ clearly } |p\rangle = |E, \alpha\rangle .$

$dp = \frac{\alpha m}{\sqrt{2mE}} dE$

so: $\int_{-\infty}^{\infty} dp |p\rangle \langle p| e^{-iEt/\hbar}$

$= \int_{-\infty}^0 dp |p\rangle \langle p| e^{-iEt/\hbar} + \int_0^{\infty} dp |p\rangle \langle p| e^{-iEt/\hbar} \quad \text{change integration variable:}$

$= - \int_0^{+\infty} dE \frac{m}{\sqrt{2mE}} |E, -1\rangle \langle E, -1| e^{-\frac{iEt}{\hbar}} + \int_0^{\infty} dE \frac{m}{\sqrt{2mE}} |E, +1\rangle \langle E, +1| e^{-\frac{iEt}{\hbar}} = \sum_{\alpha=\pm 1} \int_0^{\infty} dE \frac{m}{\sqrt{2mE}} |E, \alpha\rangle \langle E, \alpha| e^{-iEt/\hbar}$

QED.

④ Shankar
5.2.1

Let $|1\rangle$ = original ground state

$|1'\rangle$ = ground state of new well.

If well extends from $-\frac{L}{2}$ to $\frac{L}{2}$ before,
 $-L$ to L after

$$\text{then } \langle x|1\rangle = \begin{cases} \sqrt{\frac{2}{L}} \cos \frac{\pi x}{L} & |x| < \frac{L}{2} \\ 0 & |x| > \frac{L}{2} \end{cases}$$

$$\langle x|1'\rangle = \begin{cases} \sqrt{\frac{1}{L}} \cos \frac{\pi x}{2L} & |x| < L \\ 0 & |x| > L \end{cases}$$

Prob. of finding particle in $|1'\rangle$ is $|\langle 1|1'\rangle|^2$

$$= \left| \int_{-\infty}^{\infty} dx \langle 1|x\rangle \langle x|1'\rangle \right|^2 = \left| \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \frac{\sqrt{2}}{L} \cos \frac{\pi x}{L} \cos \frac{\pi x}{2L} \right|^2$$

$$\text{Let } \xi = \frac{\pi x}{2L} \\ dx = \frac{2L}{\pi} d\xi$$

$$= \left| \int_{-\pi/4}^{\pi/4} d\xi \frac{2L\sqrt{2}}{\pi L} \cos 2\xi \cos \xi \right|^2$$

$$= \frac{8}{\pi^2} \left| \int_{-\pi/4}^{\pi/4} d\xi \cos 2\xi \cos \xi \right|^2$$

$$= \frac{8}{\pi^2} \left| \frac{\sqrt{3}}{3} \right|^2 = \left(\frac{8}{3\pi} \right)^2$$

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Shankar 5.22 part (a)

(a) Show that for any normalized $|\psi\rangle$,
 $\langle\psi|H|\psi\rangle \geq E_0$, where E_0 is the
 lowest energy eigenvalue

Expand $|\psi\rangle$ in the eigenbasis of H

$$|\psi\rangle = \sum_{n=0}^{\infty} a_n |\psi_n\rangle, \quad H|\psi_n\rangle = E_n |\psi_n\rangle$$

$$\text{Then } H|\psi\rangle = \sum_{n=0}^{\infty} a_n H|\psi_n\rangle = \sum_{n=0}^{\infty} a_n E_n |\psi_n\rangle$$

Taking the inner product with $\langle\psi|$

$$\begin{aligned} \langle\psi|H|\psi\rangle &= \sum_{n=0}^{\infty} a_n E_n \langle\psi|\psi_n\rangle \\ &= \sum_{n=0}^{\infty} a_n^2 E_n = a_0^2 E_0 + a_1^2 E_1 + a_2^2 E_2 + \dots \end{aligned}$$

If E_0 is the lowest energy eigenvalue...

$$E_0 \leq E_n, \text{ for } n > 0$$

Then we can shrink all $E_n, n > 0$ to E_0
 and have the inequality

$$\langle\psi|H|\psi\rangle \geq E_0 \left(\sum_{n=0}^{\infty} a_n^2 \right) \quad \uparrow \text{ 1 since } |\psi\rangle \text{ is normalized}$$

$$\Rightarrow \langle\psi|H|\psi\rangle \geq E_0$$

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10.1.1

$$\begin{aligned}
 (1) \quad & [\Omega_1^{(1)} \otimes I^{(2)}, I^{(1)} \otimes \Lambda_2^{(2)}] \underbrace{|\omega_1\rangle \otimes |\omega_2\rangle}_{\text{test state}} \\
 &= \Omega_1^{(1)} \otimes I^{(2)} (I^{(1)} \otimes \Lambda_2^{(2)}) |\omega_1\rangle \otimes |\omega_2\rangle \\
 &\quad - (I^{(1)} \otimes \Lambda_2^{(2)}) (\Omega_1^{(1)} \otimes I^{(2)}) |\omega_1\rangle \otimes |\omega_2\rangle \\
 &= \Omega_1^{(1)} \otimes I^{(2)} (I^{(1)} |\omega_1\rangle \otimes \Lambda_2^{(2)} |\omega_2\rangle) \\
 &\quad - (I^{(1)} \otimes \Lambda_2^{(2)}) (\Omega_1^{(1)} |\omega_1\rangle \otimes I^{(2)} |\omega_2\rangle) \\
 &= (\Omega_1^{(1)} \otimes I^{(2)}) |\omega_1\rangle \otimes \Lambda_2^{(2)} |\omega_2\rangle \\
 &\quad - (I^{(1)} \otimes \Lambda_2^{(2)}) (\Omega_1^{(1)} |\omega_1\rangle \otimes |\omega_2\rangle) \\
 &= \Omega_1^{(1)} |\omega_1\rangle \otimes \Lambda_2^{(2)} |\omega_2\rangle - \Omega_1^{(1)} |\omega_1\rangle \otimes \Lambda_2^{(2)} |\omega_2\rangle \\
 &= 0 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad & (\Omega_1^{(1)} \otimes \Gamma_2^{(2)}) (\Theta_1^{(1)} \otimes \Lambda_2^{(2)}) \underbrace{|\omega_1\rangle \otimes |\omega_2\rangle}_{\text{test state}} \\
 &= (\Omega_1^{(1)} \otimes \Gamma_2^{(2)}) (\Theta_1^{(1)} |\omega_1\rangle \otimes \Lambda_2^{(2)} |\omega_2\rangle) \\
 &= \Omega_1^{(1)} \Theta_1^{(1)} |\omega_1\rangle \otimes \Gamma_2^{(2)} \Lambda_2^{(2)} |\omega_2\rangle \\
 &= (\Omega \Theta)_1^{(1)} \otimes (\Gamma \Lambda)_2^{(2)}
 \end{aligned}$$

$$(3) \quad \text{If } [\Omega_1^{(1)}, \Lambda_1^{(1)}] = \Gamma_1^{(1)}$$

$$\text{Show } [\Omega_1^{(1) \otimes (2)}, \Lambda_1^{(1) \otimes (2)}] = \Gamma_1^{(1)} \otimes I^{(2)}$$

$$((\Omega_1^{(1)} \otimes I^{(2)})(\Lambda_1^{(1)} \otimes I^{(2)}) - (\Lambda_1^{(1)} \otimes I^{(2)})(\Omega_1^{(1)} \otimes I^{(2)}))| \omega_1 \rangle \otimes | \omega_2 \rangle$$

$$= (\Omega_1^{(1)} \Lambda_1^{(1)} - \Lambda_1^{(1)} \Omega_1^{(1)}) | \omega_1 \rangle \otimes I^{(2)} | \omega_2 \rangle$$

$$= (\Gamma_1^{(1)} \otimes I^{(2)}) | \omega_1 \rangle \otimes | \omega_2 \rangle \quad \checkmark \quad \text{By the given commutation relation:}$$

$$(4) \quad (\Omega_1^{(1) \otimes (2)} + \Omega_2^{(1) \otimes (2)})^2$$

$$= (\Omega_1^{(1) \otimes (2)} + \Omega_1^{(1) \otimes (2)} \Omega_2^{(1) \otimes (2)} + \Omega_2^{(1) \otimes (2)} \Omega_1^{(1) \otimes (2)} - \Omega_2^{(1) \otimes (2)}) | \omega_1 \rangle \otimes | \omega_2 \rangle$$

$$= (\Omega_1^{(1)}) | \omega_1 \rangle \otimes I^{(2)} | \omega_2 \rangle + \Omega_1^{(1)} | \omega_1 \rangle \otimes \Omega_2^{(2)} | \omega_2 \rangle$$

$$+ \Omega_1^{(1)} | \omega_1 \rangle \otimes \Omega_2^{(2)} | \omega_2 \rangle + I^{(1)} | \omega_1 \rangle \otimes (\Omega_2^{(2)})^2$$

$$= (\Omega_1^{(2)})^{(1)} \otimes I^{(2)} + I^{(1)} \otimes (\Omega_2^{(2)})^{(2)} + 2 \Omega_1^{(1)} \otimes \Omega_2^{(2)} \quad \checkmark$$